

Accelerated Gradient Approach For Neural Network Adaptive Control Of Nonlinear Systems

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Abstract—Recent connections in the adaptive control literature to continuous-time analogues of Nesterov’s accelerated gradient method have led to the development of new real-time adaptation laws based on accelerated gradient methods. However, previous results assume the system’s uncertainties are linear-in-the-parameters (LIP). In this paper, a new NN-based accelerated gradient adaptive controller is developed to achieve trajectory tracking in general nonlinear systems subject to unstructured uncertainties that do not satisfy the LIP assumption. Higher-order accelerated gradient-based adaptation laws are developed to generate real-time estimates of both the unknown ideal output- and hidden-layer weights of a NN. A nonsmooth Lyapunov-based method is used to guarantee the closed-loop error system achieves global asymptotic tracking. Simulations are conducted to demonstrate the improved performance from the developed method. Results show the higher-order adaptation outperforms the standard gradient-based NN adaptation by 32.3% in terms of the root mean squared function approximation error.

I. INTRODUCTION

Neural networks (NNs) are universal function approximators that are capable of approximating continuous functions to a prescribed accuracy [1]–[3]. Alternative to real-time adaptation methods, numerical optimization-based methods have been used to train NNs by generating estimates of the NN weights such that a cost function is minimized over a training data set [4]. Although NN training algorithms provide a method to estimate NN weights, training is typically performed offline and then employed with open-loop control, which lacks stability guarantees. Moreover, the offline training often requires an intensive data set and significant computational resources which may require an extensive duration of time to train.

Due to the function approximation capabilities of NNs, NN-based adaptive control methods are a useful technique used to achieve control objectives (e.g., trajectory tracking) in nonlinear systems subject to unstructured uncertainties, i.e., the uncertainties do not satisfy the typical linear-in-the-parameters (LIP) assumption [5]. Results such as [6]–[10] approximate model uncertainties with a closed-loop adaptive

NN feedforward component. In these results, Lyapunov-based methods are used to develop real-time adaptation laws to estimate the weights of a NN while also providing stability guarantees.

Higher-order accelerated gradient-based optimization methods, such as Nesterov’s method [11] and the Heavy-ball method [12], have gained significant interest due to the improved transient performance and faster convergence rates compared to first-order methods [13].¹ Recent developments in [14]–[17] have made connections from discrete-time accelerated gradient methods to continuous-time analogues. Motivated by the improved convergence properties of accelerated gradient methods, insights from [14]–[17] have led to inroads for the design and analysis of accelerated gradient algorithms using Lyapunov-based methods [18]–[21]. Results in [18] and [19] develop accelerated gradient-based adaptation laws to estimate a system’s parametric uncertainties. Results in [19] apply to model-reference adaptive control models with linear dynamics and are further generalized to uncertain nonlinear systems in [20] and [21]. However, the aforementioned results in [18]–[21] are limited to systems that satisfy the LIP assumption.

In this paper, a new higher-order NN-based adaptive controller is developed for trajectory tracking in nonlinear systems subject to unstructured uncertainties. A NN model is used to approximate the uncertainties in the system. An accelerated gradient approach is used to develop higher-order adaptation laws for real-time estimation of both the output- and hidden-layer weights of a NN. The higher-order adaptation scheme is structured as two coupled first-order differential equations; the first adaptation law is used to generate auxiliary estimates of the NN weights, and these estimates are coupled to the second adaptation law which generates the true estimate of the NN weights. Consequently, the coupled structure of the developed higher-order adaptation laws impose some mathematical challenges in facilitating a Lyapunov-based stability analysis. To compensate for perturbing effects from cross-terms introduced from higher-order adaptation, the auxiliary estimates are designed into the control input. Additionally, the switching analysis in [21] is adopted in this paper to allow for activation functions with discontinuous gradients, e.g., rectified linear

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¹First-order optimization methods are recursive difference equation driven by a cost function’s gradient. Higher-order accelerated methods alter the search direction by using a weighted sum from the previous iteration to add a momentum-based term and accelerate convergence.

unit (ReLU) activation functions. A nonsmooth Lyapunov-based analysis is used to guarantee the tracking errors achieve global asymptotic tracking. Comparative simulations were conducted to demonstrate the improved performance from the developed method.

II. PROBLEM FORMULATION

A. Notation

Let $\mathbb{R}$ and $\mathbb{Z}$ denote the set of real numbers and integers, respectively. Let $\mathbb{R}_{\geq 0} \triangleq [0, \infty)$ and $\mathbb{R}_{> 0} \triangleq (0, \infty)$ denote the set of positive and strictly positive real numbers, respectively. Similarly, let $\mathbb{Z}_{\geq 0}$ and $\mathbb{Z}_{> 0}$ denote the set of positive and strictly positive integers, respectively. For $n, m \in \mathbb{Z}_{> 0}$, let $\mathbb{R}^{n \times m}$ and I_n denote the space of $n \times m$ dimensional matrices and the $n \times n$ dimensional identity matrix, respectively. Given matrices $A \in \mathbb{R}^{p \times q}$ and $B \in \mathbb{R}^{r \times s}$, the Kronecker product is denoted by $A \otimes B \in \mathbb{R}^{pr \times qs}$. Let $\text{vec}(\cdot)$ denote the vectorization operator that transforms a matrix into a column vector, i.e., for a matrix $A = [a_{ij}] \in \mathbb{R}^{n \times m}$, $\text{vec}(A) = [a_{1,1}, \dots, a_{1,m}, \dots, a_{n,1}, \dots, a_{n,m}]^T \in \mathbb{R}^{nm}$. The Frobenius norm of a matrix $A \in \mathbb{R}^{n \times m}$ is denoted by $\|A\|_F = \sqrt{\text{tr}(A^T A)}$, where $\text{tr}(\cdot)$ denotes the trace operator. The Euclidean norm of a vector $x \in \mathbb{R}^n$ is denoted by $\|x\| \triangleq \sqrt{x^T x}$. The vector space of essentially bounded Lebesgue measurable functions is denoted by $\mathcal{L}_\infty$.

B. Dynamic Model and Control Objective

Consider a control-affine nonlinear system modeled as

$$\dot{x} = f(x) + u, \quad (1)$$

where $x : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ denotes the state, $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes an unknown differentiable vector field, and $u : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ denotes a control input. The control objective is to track a user-defined desired trajectory $x_d : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ despite uncertainty of the vector field in (1). To quantify the control objective, the tracking error $e : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ is defined as

$$e \triangleq x - x_d. \quad (2)$$

The desired trajectory and its time derivative are assumed to be continuous and bounded, i.e., $x_d \in \Omega$ and $\dot{x}_d \in \mathcal{L}_\infty$, for all $t \in \mathbb{R}_{\geq 0}$, where $\Omega \subset \mathbb{R}^n$ denotes a known compact set.

III. CONTROL DEVELOPMENT

A. Neural Network Function Approximation

NN function approximation methods are well-suited for systems with unknown or unstructured uncertainties, i.e., the uncertainty does not satisfy the typical LIP assumption in adaptive control (cf., [22]–[24]). To compensate for the unknown vector field in (1), a NN-based feedforward estimate of the vector field is introduced in this section. Let the NN architecture $\Phi : \mathbb{R}^n \times \mathbb{R}^{(L+1) \times n} \times \mathbb{R}^{(n+1) \times L} \rightarrow \mathbb{R}^n$ be defined as $\Phi(x_d, W, V) \triangleq W^T \sigma(V^T \bar{x}_d)$, where $\bar{x}_d \triangleq [x_d^T, 1]^T \in \mathbb{R}^{n+1}$ denotes a concatenated state vector to account for weight biases, $W \in \mathbb{R}^{(L+1) \times n}$ denotes the output-layer weight matrix, $V \in \mathbb{R}^{(n+1) \times L}$ denotes the hidden-layer weight matrix,

$\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{L+1}$ denotes the vector of activation functions, and $L \in \mathbb{Z}_{> 0}$ denotes the user-defined number of neurons in the hidden-layer. The vector of activation functions can be composed of various activation functions, and hence, may be represented as $\sigma \triangleq [\varsigma_1, \dots, \varsigma_L, 1]^T$, where $\varsigma_i : \mathbb{R} \rightarrow \mathbb{R}$, for all $i \in \{1, \dots, L\}$, denotes a piecewise continuously differentiable activation function. Let $\mathcal{C}(\Omega)$ denote the space where $f : \Omega \rightarrow \mathbb{R}^n$ is continuous. The universal function approximation theorem in [25, Thm 3.2] states the function space of NNs are dense in $\mathcal{C}(\Omega)$. Then for any $f \in \mathcal{C}(\Omega)$ and prescribed $\bar{\varepsilon} \in \mathbb{R}_{> 0}$, there exists a constant $L \in \mathbb{Z}_{> 0}$ and ideal weight matrices $W^* \in \mathbb{R}^{(L+1) \times n}$ and $V^* \in \mathbb{R}^{(n+1) \times L}$ such that $\sup_{x_d \in \Omega} \|f(x_d) - \Phi(x_d, W^*, V^*)\| \leq \bar{\varepsilon}$. Then the unknown vector field in (1) is modeled as

$$f(x_d) = W^{*T} \sigma(V^{*T} \bar{x}_d) + \varepsilon(x_d), \quad \forall x_d \in \Omega, \quad (3)$$

where $\varepsilon : \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes the unknown bounded function approximation error. The function approximation error is bounded such that $\sup_{x_d \in \Omega} \|\varepsilon(x_d)\| \leq \bar{\varepsilon}$. To facilitate the subsequent development, the following assumption is made.

Assumption 1. [26, Assumption 9.1] The ideal NN weights can be bounded as $\|W^*\|_F \leq \bar{W}$ and $\|V^*\|_F \leq \bar{V}$, where $\bar{W}, \bar{V} \in \mathbb{R}_{> 0}$ are known constants.

B. Control Input and Weight Adaptation Laws

This section introduces the design of the controller and the higher-order accelerated gradient-based NN weight adaptation laws. The higher-order adaptation scheme is composed of two coupled first-order adaptation laws and is used to estimate the NN weights in real-time. Based on the subsequent stability analysis, the higher-order adaptation laws for the output-layer weight estimate are designed as

$$\dot{\hat{\omega}} \triangleq \text{proj} \left(\Gamma_\omega \sigma \left(\hat{V}^T \bar{x}_d \right) e^T \right), \quad (4)$$

$$\dot{\hat{W}} \triangleq -\text{proj} \left(\Gamma_W \Gamma_W \tilde{W} \right), \quad (5)$$

where $\hat{\omega} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(L+1) \times n}$ and $\hat{W} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(L+1) \times n}$ denote an auxiliary and actual estimate of W^* , respectively, $\Gamma_\omega, \Gamma_W \in \mathbb{R}^{(L+1) \times (L+1)}$ denote user-defined positive definite learning gain matrices, and $\tilde{W} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(L+1) \times n}$ is defined as

$$\tilde{W} \triangleq \hat{W} - \hat{\omega}. \quad (6)$$

The operator $\text{proj}(\cdot)$ in (4) and (5) denotes the projection operator as defined in [22, eq. E.2] and is used to ensure the weight estimates remain bounded, i.e., $\hat{\omega}(t), \hat{W}(t) \in \mathcal{W}$ for all $t \in \mathbb{R}_{\geq 0}$, where $\mathcal{W} \triangleq \{w \in \mathbb{R}^{(L+1) \times n} : \|w\|_F \leq \bar{W}\}$ denotes a known convex set and $\bar{W}$ is known by Assumption 1. The higher-order adaptation laws for the hidden-layer weight estimate are designed as

$$\dot{\hat{V}} \triangleq \text{proj} \left(\Gamma_V \bar{x}_d e^T \hat{W}^T \sigma' \left(\hat{V}^T \bar{x}_d \right) \right), \quad (7)$$

$$\dot{\hat{V}} \triangleq -\text{proj} \left(\Gamma_\nu \Gamma_V \tilde{V} \right), \quad (8)$$

where $\hat{\nu} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(n+1) \times L}$ and $\hat{V} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(n+1) \times L}$ denote an auxiliary and actual estimate of V^* , respectively, $\Gamma_\nu, \Gamma_V \in \mathbb{R}^{(n+1) \times (n+1)}$ denote user-defined positive definite learning gain matrices, and $\tilde{V} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(n+1) \times L}$ is defined as

$$\tilde{V} \triangleq \hat{V} - \hat{\nu}. \quad (9)$$

Similar to (4) and (5), the projection operator is used to ensure $\hat{\nu}(t), \hat{V}(t) \in \mathcal{V}$ for all $t \in \mathbb{R}_{\geq 0}$, where $\mathcal{V} \triangleq \{v \in \mathbb{R}^{(n+1) \times L} : \|v\|_F \leq \bar{V}\}$ denotes a convex set. Based on the subsequent stability analysis, the control input is designed as

$$u \triangleq \dot{x}_d - k_e e - k_s \text{sgn}(e) - \rho(\|e\|) e - \hat{W}^T \sigma(\hat{V}^T \bar{x}_d) + \mu, \quad (10)$$

where $k_e, k_s \in \mathbb{R}_{>0}$ denote user-defined constants, $\text{sgn}(\cdot)$ denotes the vector signum function, and $\rho : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ denotes a known strictly increasing function that satisfies $\|f(x) - f(x_d)\| \leq \rho(\|e\|) \|e\|$ for all $x \in \mathbb{R}^n$ and $x_d \in \Omega$ [27, Lem. 5]. The auxiliary term $\mu : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ in (10) is designed to cancel cross-terms introduced from higher-order adaptation in the subsequent analysis and is defined as

$$\mu \triangleq 2\tilde{W}^T \sigma(\hat{V}^T \bar{x}_d) + 2\hat{W}^T \sigma'(\hat{V}^T \bar{x}_d) \tilde{V}^T \bar{x}_d. \quad (11)$$

C. Closed-Loop Error System

To facilitate the subsequent stability analysis, the closed-loop error system is developed in this section. Let

$$\tilde{\omega}^* \triangleq W^* - \hat{\omega}, \quad (12)$$

$$\tilde{W}^* \triangleq W^* - \hat{W}, \quad (13)$$

$$\tilde{\nu}^* \triangleq V^* - \hat{\nu}, \quad (14)$$

$$\tilde{V}^* \triangleq V^* - \hat{V}, \quad (15)$$

where $\tilde{\omega}^* : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(L+1) \times n}$ and $\tilde{\nu}^* : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(n+1) \times L}$ denote the weight estimation errors between the auxiliary weight estimates and ideal weights for the output- and hidden-layers, respectively, and $\tilde{W}^* : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(L+1) \times n}$ and $\tilde{V}^* : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{(n+1) \times L}$ denote the weight estimation errors between the actual weight estimates and ideal weights for the output- and hidden-layers, respectively. Taking the time derivative of (2), adding and subtracting $f(x_d)$, and substituting in (1) and (3), the open-loop error system can be expressed as

$$\dot{e} = f(x) - f(x_d) + W^{*T} \sigma(V^{*T} \bar{x}_d) + \varepsilon(x_d) + u - \dot{x}_d. \quad (16)$$

Substituting (10) into (16) yields the closed-loop error system

$$\begin{aligned} \dot{e} = & f(x) - f(x_d) + W^{*T} \sigma(V^{*T} \bar{x}_d) + \varepsilon(x_d) \\ & - \hat{W}^T \sigma(\hat{V}^T \bar{x}_d) - k_e e - k_s \text{sgn}(e) - \rho(\|e\|) e + \mu. \end{aligned} \quad (17)$$

The first-order Taylor's series approximation of $\sigma(V^{*T} \bar{x}_d)$ yields [6, eq. (22)]

$$\begin{aligned} \sigma(V^{*T} \bar{x}_d) = & \sigma(\hat{V}^T \bar{x}_d) + \sigma'(\hat{V}^T \bar{x}_d) \tilde{V}^{*T} \bar{x}_d \\ & + \mathcal{O}^2(\tilde{V}^{*T} \bar{x}_d), \end{aligned} \quad (18)$$

where $\sigma' : \mathbb{R}^L \rightarrow \mathbb{R}^{(L+1) \times L}$ denotes the gradient of the vector of activation functions (i.e., $\sigma'(y) \triangleq \frac{\partial \sigma}{\partial z}(z) \big|_{z=y} \forall y \in \mathbb{R}^L$), and $\mathcal{O}^2(\cdot)$ denotes higher-order terms in the first-order Taylor series approximation.² Adding and subtracting the terms $W^{*T} \sigma(\hat{V}^T \bar{x}_d)$ and $\hat{W}^T \sigma'(\hat{V}^T \bar{x}_d) \tilde{V}^{*T} \bar{x}_d$ in (17) and using (18) yields the closed-loop error system

$$\begin{aligned} \dot{e} = & \chi + \hat{W}^T \sigma'(\hat{V}^T \bar{x}_d) \tilde{V}^{*T} \bar{x}_d + \tilde{W}^{*T} \sigma(\hat{V}^T \bar{x}_d) \\ & - k_e e - k_s \text{sgn}(e) - \rho(\|e\|) e + \mu, \end{aligned} \quad (19)$$

where $\chi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ denotes an auxiliary function defined as $\chi \triangleq f(x) - f(x_d) + \tilde{W}^{*T} \sigma'(\hat{V}^T \bar{x}_d) \tilde{V}^{*T} \bar{x}_d + W^{*T} \mathcal{O}^2(\tilde{V}^{*T} \bar{x}_d) + \varepsilon(x_d)$.

IV. STABILITY ANALYSIS

The closed-loop control design introduced in Section III employs a discontinuous robust sliding mode term and may have potential discontinuities in σ' depending on the choice of activation functions (e.g., ReLU activation functions). As a result, the closed-loop system is nonsmooth and does not admit classical solutions. Hence, a nonsmooth Lyapunov-based analysis is used to analyze generalized solutions of the resulting closed-loop system and ensure the tracking objective is achieved [28] (cf., [22, Thm. A.8] for LaSalle-Yoshizawa invariance principles for smooth nonautonomous systems). Specifically, the switching analysis in [10] for NN-based adaptive control is adopted to model the closed-loop system as a state-dependent switched system composed of a finite collection of smooth functions. To facilitate the subsequent analysis, let $\varrho \in \mathcal{P}$ denote a switching index, where $\mathcal{P} \subset \mathbb{Z}_{\geq 0}$ denotes a set of possible switching indices. Then the NN function approximation in (3) can be represented as $f(x_d) = W^{*T} \sigma_\varrho(V^{*T} \bar{x}_d) + \varepsilon_\varrho(x_d)$, where $x_d \mapsto \sigma_\varrho$ is smooth for each $\varrho \in \mathcal{P}$ with the corresponding function reconstruction error $\varepsilon_\varrho(x_d)$. Additionally, the functions σ', χ , and μ can be represented by the switched functions $\sigma'_\varrho, \chi_\varrho$, and μ_ϱ , respectively, where the switched functions are continuous for each $\varrho \in \mathcal{P}$. Then the closed-loop error system in (19) and the adaptation laws in (4) and (7) can be represented as

$$\begin{aligned} \dot{e} = & \chi_\varrho + \hat{W}^T \sigma'_\varrho \tilde{V}^{*T} \bar{x}_d + \tilde{W}^{*T} \hat{\sigma}_\varrho \\ & - k_e e - k_s \text{sgn}(e) - \rho(\|e\|) e + \mu_\varrho, \end{aligned} \quad (20)$$

$$\dot{\hat{\omega}} = \text{proj}(\Gamma_\omega \hat{\sigma}_\varrho e^T), \quad (21)$$

²Given suitable functions f and g , the notation $f(x) = \mathcal{O}^k(g(x))$ means that there exists constants $c, x_0 \in \mathbb{R}_{>0}$ such that $\|f(x)\| \leq c \|g(x)\|^k$ for all $x \geq x_0$.

$$\dot{\hat{v}} = \text{proj} \left(\Gamma_{\nu} \bar{x}_d e^T \hat{W}^T \hat{\sigma}'_{\varrho} \right), \quad (22)$$

for all $\varrho \in \mathcal{P}$, where the shorthand notation $\sigma_{\varrho} = \sigma_{\varrho}(V^{*T} \bar{x}_d)$, $\hat{\sigma}_{\varrho} = \sigma_{\varrho}(\hat{V}^T \bar{x}_d)$, and $\hat{\sigma}'_{\varrho} = \sigma'_{\varrho}(\hat{V}^T \bar{x}_d)$ is introduced for notional brevity. It is assumed that the bound $\sup_{x_d \in \Omega, \varrho \in \mathcal{P}} \|\varepsilon_{\varrho}(x_d)\| \leq \bar{\varepsilon}$ is satisfied. By the use of the projection algorithm in (5) and (8), the output- and hidden-layer weight estimates can be upper bounded as $\|\hat{W}(t)\|_F \leq \bar{W}$ and $\|\hat{V}(t)\|_F \leq \bar{V}$ for all $t \in \mathbb{R}_{\geq 0}$. Hence, using (13), (15), and Assumption 1, it follows that $\|\tilde{W}^*(t)\|_F \leq 2\bar{W}$ and $\|\tilde{V}^*(t)\|_F \leq 2\bar{V}$, for all $t \in \mathbb{R}_{\geq 0}$. Moreover, since $x_d, \hat{V} \in \mathcal{L}_{\infty}$ and the fact that σ'_{ϱ} and σ''_{ϱ} are continuous, it follows that σ'_{ϱ} and σ''_{ϱ} can be upper bounded by known constants for all $\varrho \in \mathcal{P}$. Then χ_{ϱ} can be upper bounded as

$$\|\chi_{\varrho}\| \leq c_1 + c_2 \|e\| + \rho(\|e\|) \|e\|, \quad \forall \varrho \in \mathcal{P}, \quad (23)$$

where $c_1, c_2 \in \mathbb{R}_{>0}$ are known constants, and $\rho(\cdot)$ was defined previously in (10).

For notational brevity, let $\Psi \in \mathbb{Z}_{>0}$ be defined as $\Psi \triangleq n + 2n(L+1) + 2L(n+1)$. Let the concatenated state be defined as $z \triangleq \left[e^T, \text{vec}(\tilde{W})^T, \text{vec}(\tilde{\omega}^*)^T, \text{vec}(\tilde{V})^T, \text{vec}(\tilde{\nu}^*)^T \right]^T \in \mathbb{R}^{\Psi}$, and let $\dot{z} = h_{\varrho}(z)$, for all $\varrho \in \mathcal{P}$, denote a collection of subsystems, where $h_{\varrho} : \mathbb{R}^{\Psi} \rightarrow \mathbb{R}^{\Psi}$. Then the corresponding switched system is represented as

$$\dot{z} = h_{p(z)}(z), \quad (24)$$

where $z : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{\Psi}$ denotes a Filippov solution to (24), $p : \mathbb{R}^{\Psi} \rightarrow \mathcal{P}$ denotes a state-dependent switching signal, and $h_{\varrho}(z)$ is defined as

$$h_{\varrho}(z) \triangleq \begin{bmatrix} g_{cl} \\ \text{vec}(\Gamma_{\omega} \hat{\sigma}_{\varrho} e^T + \text{proj}(\Gamma_{\omega} \Gamma_W \tilde{W})) \\ -\text{vec}(\text{proj}(\Gamma_{\omega} \hat{\sigma}_{\varrho} e^T)) \\ \text{vec}(\Gamma_{\nu} \bar{x}_d e^T \hat{W}^T \hat{\sigma}'_{\varrho} + \text{proj}(\Gamma_{\nu} \Gamma_V \tilde{V})) \\ -\text{vec}(\text{proj}(\Gamma_{\nu} \bar{x}_d e^T \hat{W}^T \hat{\sigma}'_{\varrho})) \end{bmatrix},$$

for all $\varrho \in \mathcal{P}$, where $g_{cl} \triangleq \chi_{\varrho} + \hat{W}^T \hat{\sigma}'_{\varrho} \tilde{V}^{*T} \bar{x}_d + \tilde{W}^{*T} \hat{\sigma}_{\varrho} - k_e e - k_s \text{sgn}(e) - \rho(\|e\|) e + \mu_{\varrho}$. In the following theorem, nonsmooth Lyapunov-based analysis techniques developed in [28] are used to establish invariance properties of (24) to ensure the tracking objective is achieved.

Theorem 1. Consider a system modeled with the dynamics in (1) and let Assumption 1 hold. Then the control input in (10) and higher-order weight adaptation laws in (4), (5), (7), and (8) ensure global asymptotic tracking in the sense that $\lim_{t \rightarrow \infty} \|e(t)\| = 0$, $\lim_{t \rightarrow \infty} \|\text{vec}(\tilde{W})\| = 0$, and $\lim_{t \rightarrow \infty} \|\text{vec}(\tilde{V})\| = 0$, provided the following sufficient gain conditions are satisfied

$$k_s > c_1, \quad k_e > c_2, \quad (25)$$

where c_1 and c_2 are known constants defined in (23).

Proof: Consider a candidate common Lyapunov function $V_L : \mathbb{R}^{\Psi} \rightarrow \mathbb{R}_{\geq 0}$ defined as

$$\begin{aligned} V_L(z) \triangleq & \frac{1}{2} e^T e + \frac{1}{2} \text{vec}(\tilde{W})^T (\Gamma_{L+1} \otimes \Gamma_{\omega}^{-1}) \text{vec}(\tilde{W}) \\ & + \frac{1}{2} \text{vec}(\tilde{\omega}^*)^T (\Gamma_{L+1} \otimes \Gamma_{\omega}^{-1}) \text{vec}(\tilde{\omega}^*) \\ & + \frac{1}{2} \text{vec}(\tilde{V})^T (\Gamma_{n+1} \otimes \Gamma_{\nu}^{-1}) \text{vec}(\tilde{V}) \\ & + \frac{1}{2} \text{vec}(\tilde{\nu}^*)^T (\Gamma_{n+1} \otimes \Gamma_{\nu}^{-1}) \text{vec}(\tilde{\nu}^*), \end{aligned} \quad (26)$$

which satisfies the inequality $\underline{\alpha}(\|z\|) \leq V_L(z) \leq \bar{\alpha}(\|z\|)$, where $\underline{\alpha}, \bar{\alpha} : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ are continuous positive definite functions. Let $F_{\varrho} : \mathbb{R}^{\Psi} \rightrightarrows \mathbb{R}^{\Psi}$ denote the Filippov regularization of (24) and be defined as $F_{\varrho} \triangleq K[h_{\varrho}](z)$, where the calculus of $K[\cdot]$ is defined in [29]. Then the generalized time derivative of (26) can be computed as $\bar{V}_L \triangleq \max_{p \in \partial V_L(z)} \max_{q \in F'_{\varrho}(z)} p^T q$ [28, Def. 3], where $F'_{\varrho}(z) \supseteq F_{\varrho}(z)$ denotes a bound on the regularization of (24), and ∂V_L denotes Clarke's generalized gradient of V_L [30, pp. 39]. Since $z \mapsto V_L$ is continuously differentiable, $\partial V_L(z) = \{\nabla V_L(z)\}$. Additionally, the time derivative of V_L exists for almost all time, i.e., $\dot{V}_L(z(t)) \stackrel{\text{a.e.}}{=} \bar{V}_L(z(t))$, where the notation $(\cdot) \stackrel{\text{a.e.}}{=}$ denotes the relation holds for almost all time.

Taking the generalized time derivative of (26), using (6), (9), and (12)–(15), performing some algebraic manipulation, and substituting in (5), (8), and (20)–(22) yields³

$$\begin{aligned} \dot{V}_L = & e^T (K[\chi_{\varrho}] - k_e e - k_s K[\text{sgn}](e) - \rho(\|e\|) e) \\ & - 2e^T \tilde{W}^T K[\hat{\sigma}_{\varrho}] - 2e^T \tilde{W}^T K[\hat{\sigma}'_{\varrho}] \tilde{V}^T \bar{x}_d + e^T K[\mu_{\varrho}] \\ & - \text{vec}(\tilde{W})^T \text{vec}(K[\text{proj}](\Gamma_W \tilde{W})) \\ & - \text{vec}(\tilde{V})^T \text{vec}(K[\text{proj}](\Gamma_V \tilde{V})). \end{aligned} \quad (27)$$

Then substituting μ_{ϱ} into (27), using (23) and the facts that $e^T K[\text{sgn}(e)] = \{\|e\|_1\}$ and $-k_s \|e\|_1 \leq -k_s \|e\|$, (27) can be bounded as

$$\begin{aligned} \dot{V}_L \stackrel{\text{a.e.}}{\leq} & -(k_e - c_2) \|e\|^2 - \lambda_W \|\tilde{W}\|_F^2 - \lambda_V \|\tilde{V}\|_F^2 \\ & - (k_s - c_1) \|e\|, \end{aligned} \quad (28)$$

where $\lambda_W \in \mathbb{R}_{>0}$ and $\lambda_V \in \mathbb{R}_{>0}$ denote the known minimum eigenvalues of Γ_W and Γ_V , respectively. Provided the sufficient gain conditions in (25) are satisfied, (28) can be upper bounded as $\dot{V}_L \stackrel{\text{a.e.}}{\leq} -\lambda \|e\|^2 - \lambda_W \|\tilde{W}\|_F^2 - \lambda_V \|\tilde{V}\|_F^2$, where $\lambda \in \mathbb{R}_{>0}$ denotes a known constant. Invoking the LaSalle-Yoshizawa theorem extension for nonsmooth systems in [28, Thm. 2], $\lim_{t \rightarrow \infty} \|e\| = 0$, $\lim_{t \rightarrow \infty} \|\text{vec}(\tilde{W})\| = 0$, and $\lim_{t \rightarrow \infty} \|\text{vec}(\tilde{V})\| = 0$. Additionally, all closed-loop signals can be shown to be bounded. ■

³Given matrices $A \in \mathbb{R}^{p \times q}$, $B \in \mathbb{R}^{q \times r}$, and $C \in \mathbb{R}^{r \times s}$, $\text{vec}(ABC) = (C^T \otimes A) \text{vec}(B)$.

V. SIMULATION

Comparative simulations were conducted on the two-state nonlinear system described in [31] to demonstrate the performance of the developed method. The unknown vector field in (1) was modeled as

$$f(x) = \begin{bmatrix} -x_1 + x_2 \\ -\frac{1}{2}x_1 - \frac{1}{2}x_2 \left(1 - (\cos(2x_1) + 2)^2\right) \end{bmatrix}, \quad (29)$$

where $x \triangleq [x_1, x_2]^T \in \mathbb{R}^2$ denotes the system state.

Two simulation experiments were performed, where each simulation was conducted for 30 s with initial condition $x(0) = [1, 0.5]^T$. The desired trajectory $x_d \triangleq [x_{d,1}, x_{d,2}]^T$ was selected as $x_d(t) = \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix}$. The NN used in each simulation had $L = 20$ neurons in the hidden-layer and contained sigmoid activation functions. The first simulation was performed with a baseline NN adaptive controller which used a typical gradient-based adaptive scheme given by

$$\dot{\hat{W}} \triangleq \Gamma_W \hat{\sigma} e^T, \quad (30)$$

$$\dot{\hat{V}} \triangleq \Gamma_V \bar{x}_d e^T \hat{W}^T \hat{\sigma}', \quad (31)$$

$$u \triangleq \dot{x}_d - \hat{W}^T \hat{\sigma} - k_e e - k_s \text{sgn}(e). \quad (32)$$

The second simulation was performed with the developed method in (4), (5), (7), (8), and (10). In both simulations, the hidden- and output-layer weight estimates were initialized randomly from the normal distribution $\mathcal{N}(0, 1)$. The parameters used in each simulation are summarized in Table I.

Table I
SIMULATION PARAMETERS

Adaptation Law	Γ_ω	Γ_ν	Γ_W	Γ_V	k_e	k_s
Standard NN Adaptive	-	-	$50I_{21}$	$50I_3$	5	0.5
Developed Method	$50I_{21}$	$50I_3$	$0.9I_{21}$	$0.9I_3$	5	0.5

Table II
RMS TRACKING ERROR, CONTROL EFFORT, AND FUNCTION APPROXIMATION ERROR

RMS	Standard NN Adaptive	Developed Method
$\bar{e}$	0.0674	0.0457
$\bar{u}$	2.6881	2.5320
$\bar{f}$	1.4546	0.9845
$\bar{e}_\tau$	0.3223	0.2169
$\bar{u}_\tau$	6.8375	5.2455
$\bar{f}_\tau$	6.3851	3.4789

Table II summarizes the performance in each simulation. In the leftmost column, $\bar{e}$, $\bar{u}$, and $\bar{f}$ denote the root mean square (RMS) tracking error, control effort, and function approximation error, respectively. The subscript $(\cdot)_\tau$ denotes the transient period before the tracking errors reach a steady-state. As shown in Figure 1, the tracking errors converge to the origin and reach steady-state after approximately 1.3 s. The RMS tracking error for the standard NN and developed

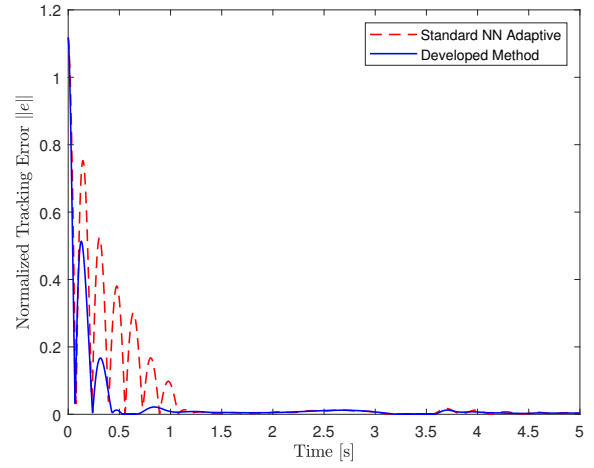


Figure 1. Evolution of the normalized tracking errors $\|e\|$. The simulation with the typical gradient-based NN adaptation scheme in (30)–(32) is shown in the dashed red line, and the simulation using the developed higher-order NN-based adaptation scheme in (4), (5), (7), (8), and (10) is shown in the solid blue line.

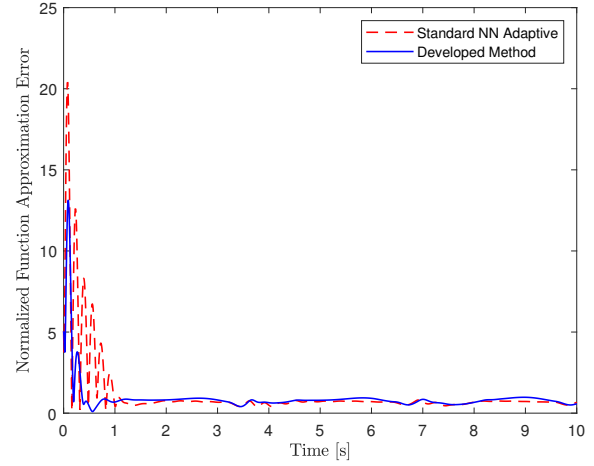


Figure 2. Evolution of the normalized function approximation errors $\|f(x_d) - \hat{f}(x_d)\|$ for each simulation.

methods were 0.0674 and 0.0457, respectively. The developed method had a 32.2% decrease in the RMS tracking error with a 5.8% decrease in the RMS control effort in comparison to the standard NN adaptive method.

Figure 2 illustrates the normalized function approximation error for each simulation. The simulation with the developed method showed a 32.3% decrease in the RMS function approximation error in comparison to the simulation with the standard NN adaptive controller. The improved transient performance from the accelerated gradient approach in the developed method is evident as the RMS function approximation error is 45.5% lower than the errors with the standard NN adaptation during the transient period from 0 to approximately 1.3 s. Additionally, the RMS tracking error and control effort showed a 32.8% and 23.3% decrease, respectively, during the transient period using the developed method. To better illustrate the improved transient performance from the developed higher-

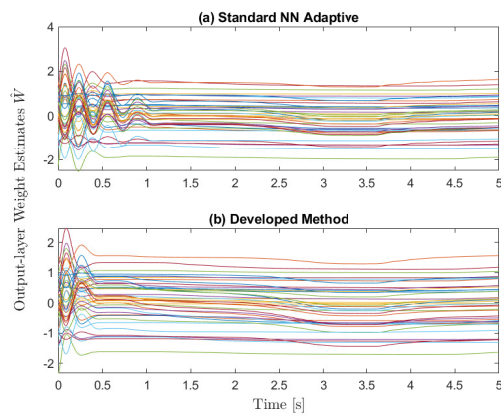


Figure 3. (a) Evolution of the output-layer weight estimates using the standard NN adaptation law in (30). (b) Evolution of the output-layer weight estimates using the developed higher-order adaptation laws in (4) and (5).

order adaptation, Figure 3 illustrates the evolution of the output-layer weight estimates. The weight estimates generated from the standard NN adaptation and the developed method are shown in Figure 3a and Figure 3b, respectively. The output-layer weight estimates from the standard NN adaptation exhibit oscillatory behavior during the transient period from approximately 0–1.3 s. However, the weight estimates from the developed method had improved transient performance as seen by the reduction of the oscillatory components in the weight estimates, and hence, resulting in improved real-time learning performance from the higher-order NN-based adaptation.

VI. CONCLUSION

Recent connections in adaptive control to continuous-time analogues of Nesterov’s accelerated gradient method have led to the development of new real-time adaptation laws based on accelerated gradient methods. Motivated by the potential improvements in transient performance, an accelerated gradient approach was used to design a higher-order NN-based adaptive control scheme for trajectory tracking of uncertain nonlinear systems. A nonsmooth Lyapunov-based analysis was performed to show the developed methods yields global asymptotic tracking. Comparative simulations were conducted on a two-state nonlinear system to demonstrate the improved performance from higher-order NN weight adaptation laws. The simulations showed the developed method outperformed the standard NN gradient adaptive scheme and had a 32.3% decrease in the RMS function approximation error.

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